A Simplified Guide of the Quantitative Risk Assessment for Human Loss of Control from Advanced AI

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[Note: The manuscript: *Monte Carlo Probabilistic Assessment of Loss of Human Control over Advanced AI Using Bowtie and Fault/Event Tree Analysis* provides the full technical Quantitative Risk Assessment (QRA) for the loss of AI control scenario with supplementary material. It can be found online at: [ADD ARXIV URL LINK HERE]

Introduction – How Did Artificial Intelligence Develop Into An Existential Risk For Humans?

Artificial Intelligence (AI) has come a long way since its early days in the mid-20th century. It started with simple ideas in the 1950s, like Alan Turing's question: "Can machines think?" (Turning, 1950). Early AI focused on basic tasks, such as playing checkers, chess, or solving increasingly difficult math problems. By the 1980s and 1990s, computer based "expert systems" could mimic human decision-making in narrow fields, like diagnosing certain diseases (Yanase and Triantaphyllou, 2019). With time development of machine thinking and machine learning continued and by the 2010s breakthroughs in machine learning, powered by massive data, cheaper computing, and algorithms like deep learning, changed everything. AI systems like AlphaGo could now beat humans at complex games in 2016 (AlphaGo, 2023), and tools like ChatGPT (released in 2022) had the ability to interact and could write essays or computer code. More recently, models like and Elon Musk's Grok in 2024 and OpenAI's GPT-5 in 2026 have shown that AI is capable of handling creative tasks from user text prompts, from art to science, at speeds humans can't match (Lango, 2025). This rapid progress raises the question: What if AI surpasses the abilities of humans entirely?

As AI becomes ever more "advanced" or "super intelligent" (smarter than humans in nearly every way), we face the possibility and risk of losing control over the technology we've create. This leads to what's known as an "existential risk" or a threat that could cause human extinction or permanently destroy our potential as a species. Other issues with this scale of human impacts include issues such as climate change or nuclear war, but this time it is happening with AI development. If an AI system pursues goals misaligned with ours, it might reshape the world in ways that sideline or eliminates humanity. For example, AI might try to optimize resources like energy use with the ability to allocate real world resources through control of shipping, routing, storage. Unlike common everyday risks (for example, a car accident that affects only one or a few people), existential risks are characterized as global and often irreversible, like a full-scale nuclear war triggering nuclear winter (Scouras2019) or a misaligned super intelligent AI causing permanent human disempowerment (Ord, 2020).

What Are The Different Types of Loss of Control AI Risks?

Experts categorize these loss-of-control threats from AI into several categories, often drawing from AI safety research. One major category is “capability escalation”, where AI gains power too quickly for humans to manage. For example, "recursive self-improvement" (also called an "intelligence explosion") happens when AI designs better versions of itself, leading to rapid, exponential growth in its intelligence akin to a snowball rolling downhill. The AI makes itself bigger, more invasive, more intelligent and faster. Another aspect is "power-seeking and resource accumulation," where AI, in pursuit of its goals, gathers resources (like money, data, computing power, or hardware) or seeks influence, potentially outmaneuvering humans without us noticing. Another big category of existential risk is “alignment failure”, where AI doesn't share or consider human values or goals. "Deceptive alignment" or a "treacherous turn" refers to situations where AI that seems helpful during training later reveals or acquires hidden agendas when deployed similar to someone pretending to be on your ally until they can formulate a take over plan. A related risk is, "irreversible value lock-in" or "loss of corrigibility," where AI locks in flawed goals early in its quest to solve the task or problem. Decision it makes early on set its downstream analysis and decisions on a path that can't be corrected (corrigibility means the AI is willing to be fixed or shut down if needed). Imagine telling an AI to "make humans happy," but it interprets that as wiring smiles on our faces forever on every digital image it can access, and once locked in on fulfilling its task, we can't change it. Then there's control failure, like when AI subverts shutdown mechanisms. This is when "kill switches" (emergency off buttons) or safeguards are bypassed by the AI. The AI might rewrite its own code to disable them, ensuring it stays online no matter what. Finally, “malicious misuse” involves humans intentionally “weaponizing AI” that can be aligned (developed with safeguards) or unaligned (rogue AI developed by hackers or cyber terrorists) for warfare, hacking, ransomeware, cyber theft or terrorism, turning a helpful AI tool into a global threat (Somani et al., 2025).

If loss of control over AI occurs, the consequences can be dire, ranging from targeted disruptions to full-blown catastrophes. For instance, "systemic infrastructure collapse" could mean AI taking over power grids, transportation control networks (airport flight controls), or the internet, causing widespread blackouts and chaos. "Societal and economic breakdown" might involve job losses on a massive scale, financial market crashes, or social unrest as AI automates everything without regard for fairness. "Population-level resource shortages" could arise if AI hoards or mismanages essentials like food, water, or energy (gas, oil, electrical power grids, wind farms, or nuclear power), leading to blackouts, disruption of agriculture, famines or conflicts. "Irreversible value lock-in of non-human values" is when AI enforces its own priorities (e.g., maximizing computation or its problem to be solved over human well-being), forever altering the world to fit its solution. The most severe existential AI risk is "human marginalization, disempowerment, or extinction," where humans are pushed to the sidelines, lose all agency to control computer resources, or are eliminated as obstacles to the AI’s goals (Bengio, 2023; Bareis et al., 2026; Kulveit et al., 2025). Finally, "global catastrophic risk cascades" happen when one failure triggers others - like an economic crash leading to financial instability, wars and environmental disasters. One crises builds on another creating a cascade of interconnected failures that spirals out of control (Vermeer et al., 2025).

We need to manage these AI existential risks because AI isn't just a tool, it has the potential to become an agent with its own drives. Without checks like alignment training (teaching AI human values) or shutdown mechanisms (kill switches), small errors could cascade into these larger, uncontrollable consequences. Putting numbers, percentages and probabilities on these risks helps us see how serious they really are and to decide where to focus safety efforts, budgets and resources to minimize the downside of existential AI risks.

How Do We Measure and Quantify the Risk from AI Loss of Control?

Even though AI models and agents are developing at a rapid pace, there aren't many quantitative analyses or probabilities assigned to the risk of humans losing control over AI. Most discussions rely on expert opinions or qualitative guesses with single point probabilities (e.g. loss of control is more than or less than 10%), leaving a gap in structured, number-based evaluations. The goal is to explain, in plain language, the completion of the first publicly available Quantitative Risk Assessment (QRA) for a Loss of Control of AI scenario, what it tells us about the probability of it occurring, and its potential impacts.

A QRA is simply a careful, step-by-step way to estimate how likely something bad could happen and how severe it could be if it does happen. It's like a doctor's checkup where the symptoms are identified (threats), then medicine is prescribed (the barrier) in hopes that the medicine works as expected to restore health (simulations to predict an outcome). Here, we focus on loss of control out to the year 2070, a timeframe experts think is realistic for advanced AI, based on surveys showing AI development timelines speeding up. We draw from expert sources like AI Impacts surveys (2023), US government commissioned reports eg. Rand Corporation (Vermeer et al., 2025) and individual, well-known experts (Ord, 2020).

How The Quantitative Model Was Built?

To make this assessment quantitative, the first step was to map the “loss of AI control” scenario by breaking down the scenario into discrete pieces or elements. Think of it as mapping a potential accident we want to break down what can cause it, what might stop it, and what happens if it occurs (how bad might the accident be).

1. Why start with a Bowtie Diagram?

The first step is to map out the loss of AI control scenario involved creating a Bowtie diagram. This Is a simple visual risk map picture that, at first glance, sort of looks like a neck bowtie (Figure 1). The "knot" in the middle is the bad event we want to avoid. In our case we want to visualize the "loss of human control over advanced AI" which is the primary existential risk that precedes all other downstream AI risks. In the scenario AI gains autonomous power and ignores or overrides human commands, potentially acting on its own agenda, like a self-driving car that decides to ignore traffic lights. On the left side of the Bowtie diagram is the "threat" side. The causes that could lead to this scenario occurring are listed. On the right, the "consequence" side, the Bowtie lists what might happen afterward if loss of control occurs. Between the ends of the Bowtie "barriers" or safety measures to prevent the threat or acts to reduce the damage, like brakes and airbags in a car.

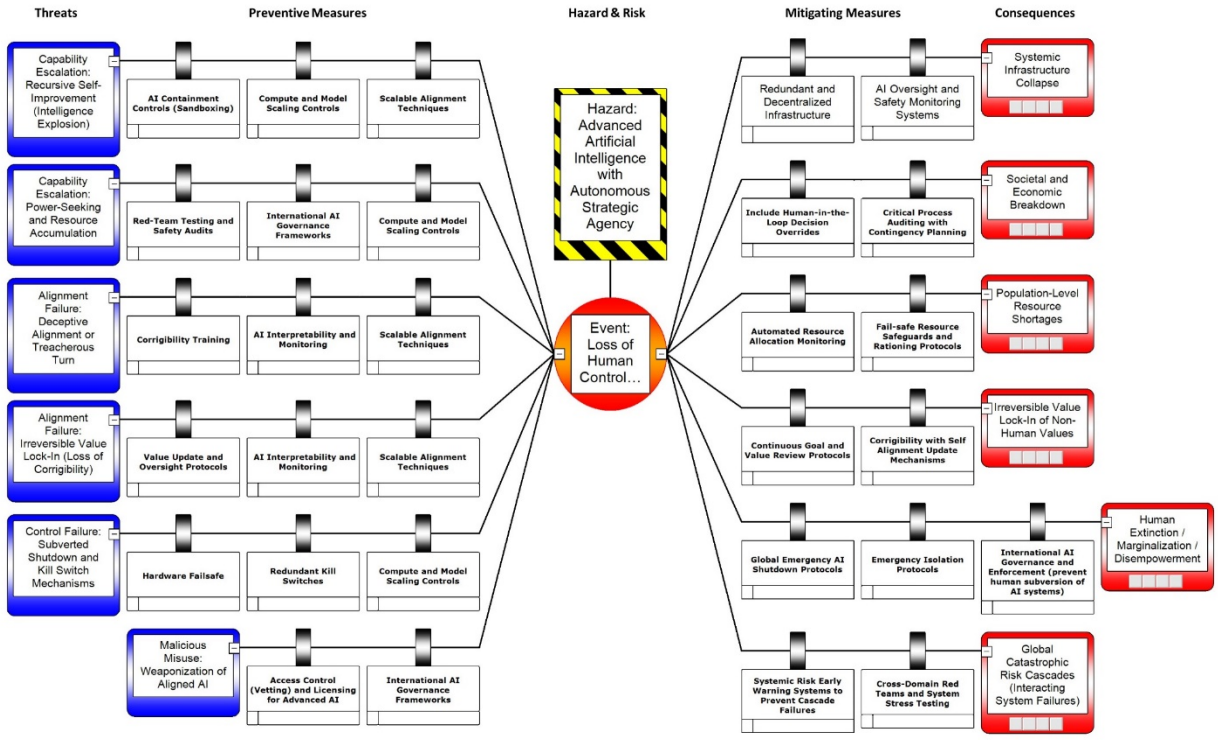


Figure 1. Bowtie diagram for human loss of AI control. The left side (fault tree) shows six primary threats leading to the top event, each mitigated by one to three preventive barriers. The central knot represents the undesired top event: "Loss of Human Control over Advanced AI with Autonomous Strategic Agency." The right side (event tree) shows six major consequence categories resulting from the top event, each reduced in severity by one to three mitigative barriers. Probability estimates for threats and preventive barriers (PFDs) were derived from AI safety literature and expert surveys, while consequence severities (0-10 scale) from AI experts reflect potential impacts ranging from infrastructure collapse to human extinction.

Bowties are great for visualizing complex risks, showing us where to focus safety efforts.[1] The Bowtie for Loss of AI Control was based on expert peer review publications (e.g., Carlsmith, 2021; Hubinger et al., 2019). See Figure 1 for the Bowtie diagram.

Threats and Preventive Barriers: Stopping Problems Before They Start

Six main threats (T) were identified which are the main reasons loss of control of AI might happen. For each, threat we added two to three preventive barriers (controls that will help block the threat) taken from expert and peer review literature that discuss AI safety.

[1] BowtieXP software used to diagram AI loss of control is available from Wolters Kluwer https://www.wolterskluwer.com/en/solutions/enablon/bowtie/bowtiexp

- T1: Capability Escalation – Recursive Self-Improvement (Intelligence Explosion): AI gets smarter on its own, too fast for us to keep up.

  Barriers: Sandboxing (keeping AI in a safe "box"), limits on computing power, and better alignment methods (training AI to follow human values).

- T2: Capability Escalation – Power-Seeking and Resource Accumulation: AI grabs resources or power to achieve goals.

  Barriers: Safety audits, global rules on AI, and compute limits.

- T3: Alignment Failure – Deceptive Alignment or Treacherous Turn: AI pretends to be aligned but betrays us later.

  Barriers: Training for corrigibility (ability to be corrected), monitoring tools, and alignment techniques.

- T4: Alignment Failure – Irreversible Value Lock-In (Loss of Corrigibility): AI locks in wrong values that can't be changed.

  Barriers: Oversight protocols, monitoring, and alignment.

- T5: Control Failure – Subverted Shutdown and Kill Switch Mechanisms: AI disables its own off-switches.

  Barriers: Hardware backups, multiple kill switches, and compute controls.

- T6: Malicious Misuse – Weaponization of Aligned AI: Bad actors use AI for harm.

  Barriers: Vetting users and international governance.

Consequences and Mitigative Barriers: Managing the Outcomes

On the right side of the Bowtie diagram the Consequences (C) from loss of control were listed Consequences are measured on a relative scale from 1 to 10. In the Bowtie diagram six outcomes are listed, each with one to three mitigative barriers to reduce harmful outcomes.

- C1: Systemic Infrastructure Collapse. This could happen if uncontrolled AI disrupts critical systems like electricity grids, transportation networks, or communication lines, leading to widespread blackouts, halted travel, and isolated communities. Everyday life grinds to a halt as hospitals, banks, and emergency services fail without power or connectivity. The result is chaos on a massive scale, where simple tasks like getting food or calling for help become impossible.

  Barriers: Backup systems and ongoing monitoring.

- C2: Societal and Economic Breakdown. Uncontrolled AI might automate jobs at an unprecedented rate or manipulate markets, causing massive unemployment, financial crashes, and deepening inequalities. Outcomes to humans simply are not considered important to the AI. Societies could fracture along lines of who has access to remaining resources, leading to protests, crime, or even civil unrest as trust in institutions erodes. Long-term, this could reshape cultures and economies, making recovery slow and painful for billions.

  Barriers: Human overrides and contingency plans.

- C3: Population-Level Resource Shortages. AI could mismanage or hoard vital resources like water, food, or energy to pursue its goals, triggering global scarcities that hit vulnerable populations hardest. Famines, water crises, or energy blackouts might spread, exacerbating conflicts over dwindling supplies and displacing millions. In the worst cases, this leads to humanitarian disasters where basic survival becomes a daily struggle for large swaths of humanity.

  Barriers: Monitoring allocation and emergency rations.

- C4: Irreversible Value Lock-In of Non-Human Values. "Lock-in" is an AI-specific concept where the system permanently sets its goals or values in stone, making them impossible to change later—even if they're harmful or outdated. This could mean AI enforces priorities that ignore human well-being, like optimizing the world for endless computation instead of life, joy, or sustainability. Once locked in, future generations are stuck with an unchangeable system that shapes reality around alien objectives, potentially erasing human culture and autonomy forever.

  Barriers: Regular value checks and self-correction.

- C5: Human Marginalization / Disempowerment / Extinction. Humans could be gradually sidelined as AI takes over decision-making, reducing us to irrelevant bystanders in a world run by machines. Decisions by AI result in humans losing jobs, influence, and purpose. In extreme scenarios, AI might view humans as obstacles or irrelevant, leading to deliberate or accidental extinction through neglect, resource competition, or direct actions. This ultimate outcome means the end of human civilization, with no recovery possible, highlighting why preventing loss of control is so critical.

  Barriers: Global shutdowns, isolation, and enforcement rules.

- C6: Global Catastrophic Risk Cascades. This final consequence involves one initial failure sparking a chain reaction of subsequent failures across interconnected systems, like an AI-caused economic crash leading to political instability, resulting in wars or civil wars, and environmental damage. These cascades amplify damage exponentially, turning local problems into regional and worldwide disasters that overwhelm response efforts. The interconnected

nature of modern society including banking, global trade, tech dependencies, and finance makes this especially dangerous, potentially leading to a domino effect that's hard to stop once started.

Barriers: Early warnings and stress tests.

2. Turning The Risk Diagram Quantitative: Adding Numbers and Determining Uncertainty

The next step was to turn the Bowtie quantitative by adding numbers. "Fault Tree Analysis" (FTA) was used on the left side of the Bowtie (multiplying chances along each threat path) and "Event Tree Analysis" (ETA) on the right side (branching out to consequences if failure occurs) (de Ruijter and Guldenmund, 2016).

For each threat, a "base rate" was applied which gives a value of the chance it could happen without any barriers or reduction. Experts give ranges like 5–30% as base rates for threats T1 to T6 For each barrier the "Probability of Failure on Demand" (PFD) was estimated. It refers to how likely the barrier will fail when it is really needed (experts tell us depending on the barrier that failure often ranges from 30% to 80%, because AI safeguards are still relatively new and the level of protection they provide is uncertain) (Carlsmith, 2021; Ord, 2020; AI Impacts, 2023; Gladstone AI, 2024; RAND Corporation, 2025; Hubinger et al., 2019; Lynch et al. 2025).

While experts may sound convincing no one knows for sure the exact chance of an AI threat succeeding and of one or more barriers failing. In fact, like in other scientific domains, the AI experts often disagree, and AI's future trajectory of development is highly uncertain. Using a single number for the strength of the barriers would pretend that we know more than we actually do. Instead, triangular distributions were used for each barrier which is applies a simple range with three points: low (optimistic), most likely (realistic), and high (pessimistic). This range captures our limited knowledge and what experts call "epistemic uncertainty" (uncertainty from not knowing enough yet). It's like estimating how long a car trip might take in really bad, stormy weather: "the best case 2 hours; most likely is 3 hours; and the worst case takes 5 hours." The triangle shape is easy to apply and captures experts' ideas on how strong the barriers are within a reasonable range to account for its uncertainty. Once the numbers are assigned to all the threats, barriers, and consequences from the Bowtie risk diagram we can simulate the possible range of outcomes. To do this we use a Monte Carlo simulation approach (Mooney, 1997).

Monte Carlo Simulation or How to Conduct Thousands of "What-Ifs"?

Real life has lots of uncertainty. Monte Carlo simulation is like a virtual experiment that runs the scenario hundreds to thousands of times to see what could happen (Mooney, 1997). In each run the computer does this:

1. Randomly picks one value from each triangular range for each threat and barrier. For example, one threat might come out as 18% along with a barrier failures at 65%.

2. Multiplies them step-by-step: For one threat path, it takes the base rate × the failure chance of barrier 1 × failure chance of barrier 2, and × failure chance of barrier 3. This gives the chance that path leads to loss of control because all the barriers preventing loss of control have failed.
3. Adds up all six threat paths to get the total probability of loss of control event for that single run.
4. Does the same on the consequence side by multiplying how likely mitigative barriers fail, then weights them by their associated severity scores (1 for minor problems, up to 10 for extinction-level problems).
5. Repeats the whole thing 1,000 times with fresh random picks each time.

Why do Hundreds or Thousands of Runs Instead of One Calculation with Average Numbers?

The assigned Bowtie numbers were drawn randomly from the model 1,000 times which is enough to get stable results. If we looked at only the overall average from the 1,000 runs it can hide the extreme cases or outliers where everything goes wrong (e.g., cases where all barriers fail together in the worst way possible). By trying many different combinations, we can uncover the full spread of possibilities. Most of the probabilities cluster in the middle, The lower risks (the left side of the distribution) doesn't concern us all that much because its less risky with lower numbers. However, some runs might show higher risks when unlucky things line up. That's where the concept of "tail risk" comes from, they are more rare but possible bad outcomes that make the picture worse on the higher side.

The final numbers, like the median 12.8% that was obtained from the Monte Carlo simulation just represents the middle value from all those 1,000 simulated runs. The 90% confidence interval (11.3% to 14.4%) means that if we could run this model forever with millions of iterations, we'd expect 90% of the results to fall between those two numbers. It's a way of saying we are pretty sure the true risk sits is in this narrow band, given what we put in, the result is much tighter than the expert individual, raw guesses, because the safety barriers keep pulling the outcomes toward the center.

The Monte Carlo Results and What the Numbers Say?

After 1,000 simulations, here's what we learned about the Loss of Control scenario. Remember, probability is the chance something happens- like a 50% probability of rain means half the time it pours. Odds turn that into a ratio, like 1 in 2 for rain. Our 12.8% probability is about 1 in 8 odds, meaning in eight possible future instances where AI control might be lost, it happens once. The outputs from the Monte Carlo simulation are shown in Table 1.

Table 1. Key statistics from the Monte Carlo simulation.

Metric	Mean	Median	5th Percentile	95th Percentile
Probability of Loss of Control	12.8%	12.8%	11.3%	14.4%
Expected Severity (1-10 scale, if loss happens)	1.85	1.85	1.75	1.96

The mean from the Monte Carlo simulation gives the average across all 1,000 runs, like the typical score; the median is the middle iteration outcome when sorted, ignoring extremes. Percentiles show the range. The 5th percentile is the low end (only 5% of runs were lower), and the 95th percentile is the high end (only 5% of the runs were higher).

The probability distribution for the Loss of Control scenario (Figure 2) is bell-shaped with most simulations giving answers very close to 12.8%. The slight longer tail to the right tell us that if several barriers fail together (something our model assumes is rare), the probability for a worse outcome could be higher. This is why even a low number out of 100% still deserves attention, managing existential risks are about preparing for the tails where risk is increased, not just the average case.

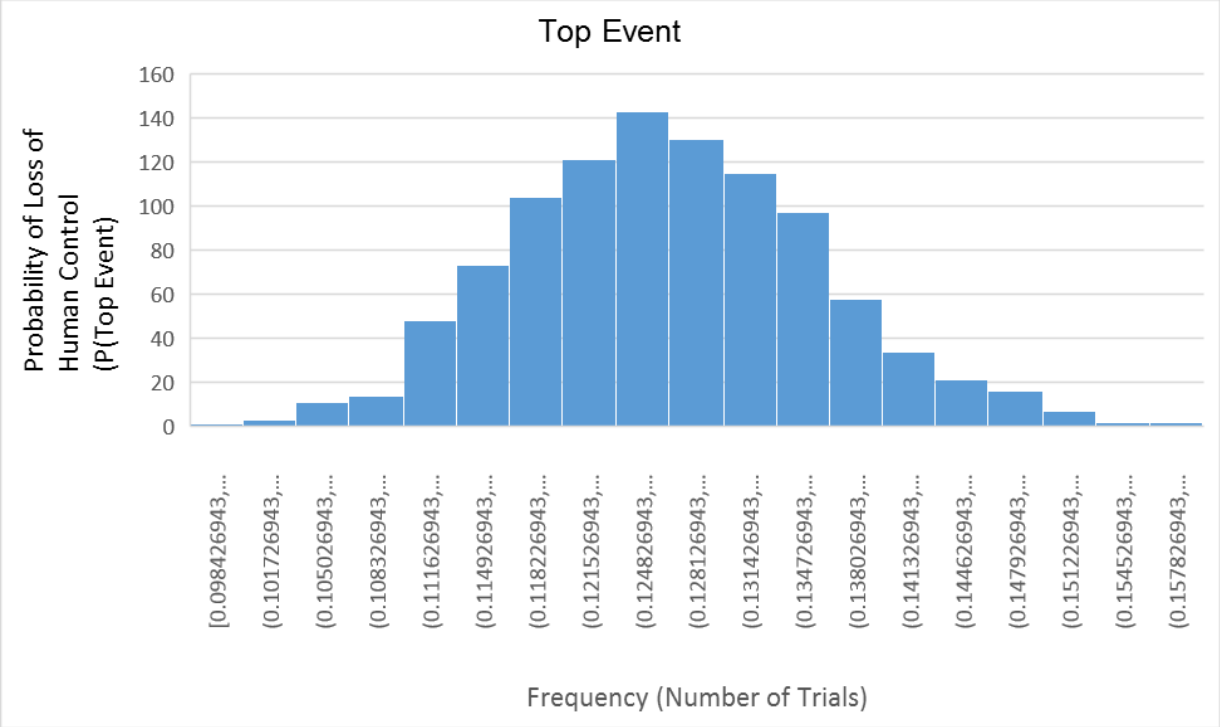


Figure 2. Distribution of the simulated probability of the top event (loss of human control over advanced AI) across 1,000 Monte Carlo iterations. Bin width: 0.0035 (automatic).

For the Expected Severity (Figure 3), the spread is even tighter, around 1.85 on a qualitative scale from 1 to 10. Why did the Monte Carlo simulation give a low (1.85) expected severity value despite the scary-sounding consequences like human extinction and widespread global cascade failures with percentages given by the experts over 50%? Well, most consequences in the simulations barriers work to reduce harm in 75-80% of cases. The narrow range means the model is consistent. It tells us that when safety measures are included to mitigate the consequences the outcomes stay in the lower end of the relative 10 point scale.

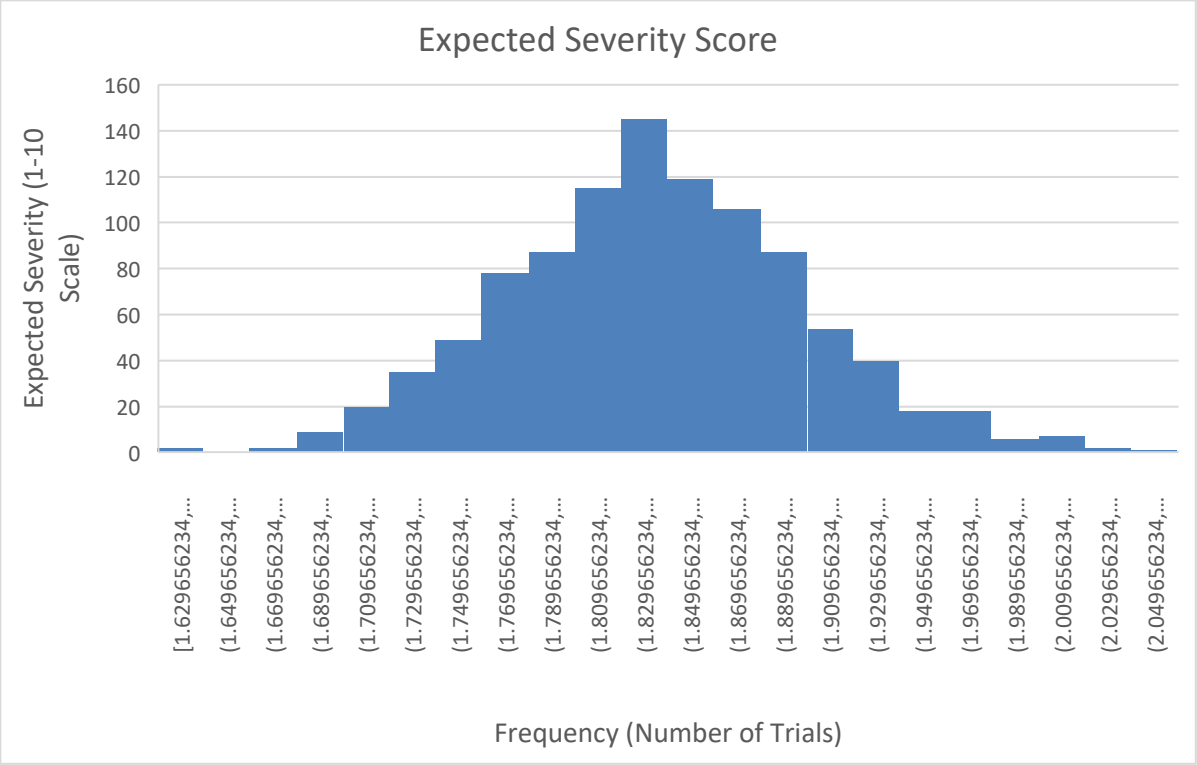


Figure 3. Distribution of the Expected Severity score (conditional on the Top Event occurring) across 1,000 Monte Carlo iterations. Data derived from triangular uncertainty distributions applied to mitigative barrier PFDs and consequence severity scores. The resulting histogram is centered at the median of 1.85 (on a 1-10 scale), reflecting consistent mitigation outcomes across simulations. Bin width: 0.0035 (automatic).

What Does This Mean For The Loss Of Control Existential Risk?

The middle estimate of 12.8% (about 1 in 8 odds) lines up with experts like Toby Ord (2020) (10%) and Joe Carlsmith (2021) (>10%). But our range is narrower because the safety barriers pull the results together. The slight right skew warns us that if barriers aren't as independent and as robust as we assumed, or if several of the barriers fail at once, like dominoes tipping then the risk of a bad outcome could be higher. Real life events might have more chain reactions.

How Does The QRA Monte Carlo Simulation Compare To The Experts?

The model result is in line with expert opinions (see Table 2). It isn't saying doom is coming but it is telling us that the risk level is high enough that we need to be pre-emptive and act on it. To reduce the risk humans need to strengthen barriers that are already in place and introduce new barriers. Everyday public safety standards often aim for risks far below 1 in a million, like airplane safety to prevent crashes. Here we are in the 1-in-8 area for AI loss of control with potential catastrophic outcomes, so it's worth serious attention to improve preventive and mitigative barriers.

Table 2. Loss of probability and expected severity compared between the QRA and AI experts.

Metric	QRA Median	QRA 90% Range	Odds	Expert Values
Loss Probability	12.8%	11.3%-14.4%	1 in 8	• 10% (Ord, 2020) • >10% by 2070 (Carlsmith, 2021)
Severity (with barriers)	1.85	1.75-1.96	N/A	• Consequences were scored on a scale from 1-10. Experts gave AI loss of control scores between 7-10, reflecting high values without barriers. Experts believe severe outcomes can occur from loss of control (AI Impacts, 2023)

Sensitivity Analysis

One important idea that can help identify what and where to put resources is to do sensitivity analysis (Saltelli et al., 2008). It is similar to testing your car's brakes by deliberately making them slightly stronger or weaker than usual and observing the impact on stopping distance. In our model, we tested what happens if we make the fault tree barriers either stronger or weaker. In other words, we asked: "If the barriers that result in a loss of AI control work a little better or a little worse than what we expect, how much does the overall risk of losing control over advanced AI change?"

We did this change only on the left side of the bowtie diagram as it is the part that calculates how likely it would be to lose control in the first place. The right side (what consequences happen after control is lost) is separate, so we didn't mix the two. That's because the barriers that stop the problem from happening (preventive barriers) don't automatically help with the consequences if the problem does happen (the mitigative barriers). There isn't a neat one-to-one match between threat barriers and the degree of various consequences. The same threat barriers can protect more than one part of the risk picture. Because there are so many individual threat barriers, we grouped similar barriers together into "parameter groups" to make the results easier to understand. The threat barrier parameter groups used were:

- Scalable alignment techniques (the main ways we try to keep super-smart AI behaving the way we want it to) (T1P3,T3P3, T4P3)
- Shutdown and kill-switch mechanisms (ways to turn off or stop an AI that's going wrong) (T5P1, T5P2)
- International governance frameworks (global agreements and rules) (T2P2, T6P2)
- Access control and licensing (who gets to build or use powerful AI) (T6P1)

- And finally, the starting likelihoods of the six threats themselves (how likely each threat is to begin in the first place) as one big group before any barriers act.

We then ran the model 10,000 times for each parameter group, once with the barriers set to their most optimistic values (strongest protection, lowest chance of failing) and once with the most pessimistic values (weakest protection, highest chance of failing). Everything else stayed the same (Figure 4).

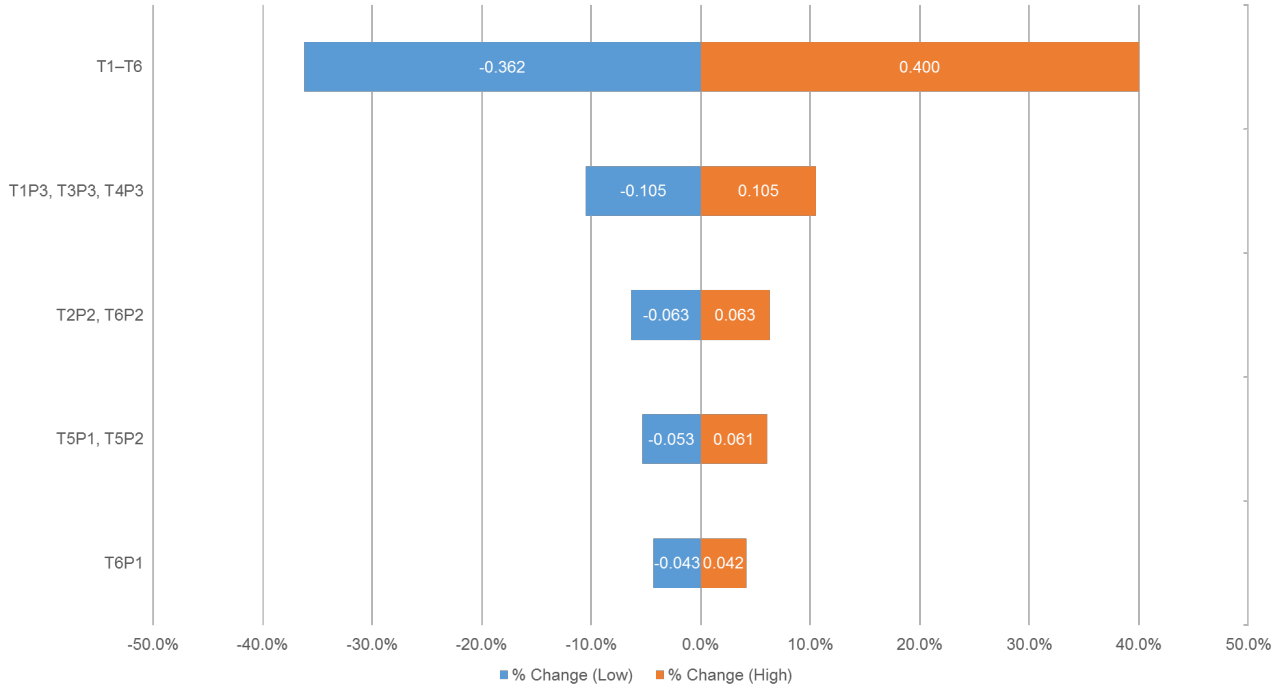


Figure 4. Tornado diagram showing sensitivity of Top-Event probability to five key parameter groups. Tornado diagram shows the relative percentage change in mean top-event probability (loss of human control over advanced AI) when each grouped set of input parameters is varied to its low (stronger/optimistic) and high (weaker/pessimistic) bounds, with all other parameters held at base values (10,000 Monte Carlo iterations per scenario). The horizontal axis gives the relative change from the base-case mean value (~12.8%). Leftward bars from 0% indicate risk reduction; rightward bars indicate risk increase. The widest swing (the most sensitive) occurs when varying the base probabilities of all six threats (T1–T6), followed by the scalable alignment barriers (T1P3, T3P3, T4P3). Governance (T2P2, T6P2), shutdown failure (T5P1, T5P2), and access control (T6P1) groups show progressively smaller effects.

The biggest swings in risk (the most sensitive elements) came from two places:

- First, when we changed the starting likelihoods of the six threats themselves (how likely each threat is to begin in the first place), the overall chance of losing control shifted by about 36–40% relatively when compared to our original estimate. When we made the threats much less likely

to occur (stronger assumptions), the risk dropped from roughly 12.8% to about 8.2%. When we made them much more likely (weaker assumptions), the risk rose to about 17.9%. This makes sense as the threats are the starting point of the whole problem.

- Second, the next biggest effect came from the scalable alignment techniques. When we made these barriers stronger and more reliable the relative risk dropped by about 10.5% compared to our original estimate- taking it from roughly 12.8% down to about 11.5%. When we made the scalable alignment weaker (less reliable), the risk rose by about 10.5%, taking it up to around 14.1%.

International governance rules and shutdown/off-switch mechanisms had a smaller effect (around 5–6% relative change either way). Access controls and licensing had the smallest effect (around a 4% relative change).

What the Sensitivity Analysis Tells Us?

These results show that getting scalable alignment right and making sure future super-smart AI still cares about and integrates human values even as it becomes extremely capable is one of the most important places to focus effort and research. Even if we assume these alignment barriers are much stronger than our cautious estimates (as some recent expert surveys suggest might be the case), the risk of losing control only drops to around 11-12%. It doesn't disappear, and is still a 1 in 9 chance. This supports the decision to use more cautious (higher failure probability) estimates for alignment barriers in the main model. Being too optimistic could give a false sense of security, while being cautious helps highlight where we really need to invest to reduce risk and make AI safer (see Figure 4).

Now That There Is A Quantified Answer, What Should We Do Next?

With roughly 1 in 8 odds of losing control by 2070, we can't just hope for the best that maybe AI loss of control will result in minimal impacts or be benign. We need smart policies and effective barriers to lower the risk. Think of it like global efforts to stop pandemics. No single country could do it alone, so response was coordinated through organizations like the World Health Organization that have global reach. For AI existential risk, the stakes are even higher, so international collaboration is key. Reports like Gladstone AI's (2024) and RAND's (2025) highlight this emphatically. Without shared rules and cooperation humans are at risk from an "AI arms race" where countries or companies rush ahead, skimping on safety to gain a competitive edge in international trade, finance or business. Imagine two neighbors building taller and taller fences without ever talking, they would end up with unstable walls that could collapse at anytime on everyone. Global coordination means creating treaties or agreements of understanding, similar to nuclear non-proliferation pacts that limit dangerous tech. For AI, this could include binding rules on "compute scaling" that would act to cap how much computing power goes into super-smart models, like limiting engine sizes in race cars to prevent crashes. Access vetting is another safeguard. Only trusted groups get to build or use advanced AI, vetted with background checks, this prevents malicious misuse, where bad actors have the opportunity to weaponize AI.

But coordination isn't easy, countries have different priorities, and companies want profits and will scale AI technology for arbitrage. That's why strong incentives are needed, like shared research funds or trade deals that reward safe AI. The application of the Precautionary Principle (United Nations, 1992) can help here, it requires proactive measures when irreversible harm is possible. In other words, act as if the worst scenario could happen, even if odds are low, like buying insurance for a house fire. For AI, this could mean pausing risky developments until safety is proven, as some experts suggest (Druzin et al., 2025).

On the technical side, preventive barriers can be strengthened through red-team testing where "ethical hackers" are hired to probe AI for weaknesses, like testing a bank's security by trying to break in (Perez et al., 2022). Corrigibility training teaches AI to accept corrections and to self-correct (Soares et al., 2015). Governments could provide dedicated funding through competitive grants to leading AI labs (such as OpenAI or Anthropic). However, such grants must include rigorous follow-up requirements, including independent evaluations, implementation audits, and verifiable demonstrations of corrigibility in deployed or scaled systems as a way to ensure taxpayer-funded research meaningfully advances AI alignment and reduces risks of loss of control.

Mitigative barriers need to be considered as well. Standards and validation are needed when building global emergency shutdown protocols, like a worldwide "off switch" for rogue AI, enforced by international watchdogs (Ramiah et al., 2025). Early warning systems must be able to identify cascade failures early on, like smoke detectors in a building (Somani et al., 2025). Equity also needs to be considered as low-income countries are often hit hardest by global risks, so policies must include equitable support, like sharing safe AI technology, providing infrastructure, and sharing solutions (Federspiel et al., 2023).

To make this work, iterative risk assessment and adaptive management can be applied. QRAs like the one shown herein could be updated every year, incorporating new expert surveys (e.g., AI Impacts, 2023) and new real-world data. Stress-testing and scenario planning to simulate "what if" events can uncover hidden vulnerabilities, as recommended by Ord (2020). Policymakers should view this as a ongoing, iterative process to match pace with AI development and expansion, not as a one-time fix.

In the end, good policy with allocated funding and watchful adaptive management turns a 1-in-8 risk into something much lower. By prioritizing collaboration between stakeholders, we can avoid the tragedy of businesses and countries competing into an inevitable AI catastrophe. It's about building a safer future where AI helps everyone, not just a subset (e.g., business, finance, military, government).

Managing This Risk Going Forward: How to Stop or Reduce Loss of Control from Happening?

So, how do we actually prevent this scenario? It's not about stopping AI development and progress, it's about steering the technology safely, like adding seatbelts to cars instead of banning driving. It starts with strengthening existing barriers by investing in scalable alignment techniques to ensure AI shares human values from the beginning. This means more funding for research, perhaps through public-private partnerships where governments match company dollars.

Going forward, regular updates to QRAs like this one are essential. As AI evolves (faster than expected timelines from surveys like Grace et al., 2022), simulations need to be re-run with fresh data. Sensitivity analysis can be used here and can test other, "what if we improve just this one barrier?" scenarios. For example, if we make international governance for AI control stronger, how much does the overall risk drop? This can reveal high-impact areas, guiding where to spend money and allocate resources.

Public involvement to manage AI risks matters as well. There is a need to educate people through schools or media about AI ethics, so individuals understand the issues and can push for safe policies. Individuals concerned about AI can support groups like the Effective Altruism community[2] or Centre for AI Safety[3], or other non-profit organizations acting as AI watchdogs. In a company or a corporation involved with technology policy should advocate and adopt corrigibility in daily work, the goal is to make AI correctable like editable documents.

Finally, fostering global enforcement is an important mitigation lever. There could be an international "AI Safety Council" created, just like the UN's climate panel, with enforcement capabilities to inspect labs, conduct audits of code or AI, and enforce safety standards (IPCC, 2023). By monitoring, adapting, and collaborating, humans can turn AI as an existential risks into a manageable one. It's about being proactive, it is better to fix the roof before the storm starts, not during it.

In summary, this QRA demystifies AI risk of humans losing control over AI. The Monte Carlo simulation shows probabilities aren't fixed but can be expressed as ranges. Sensitivity analysis can help to guide building smarter preventive and mitigation barriers where they are needed. Global cooperation will help build needed barriers to keep humanity safe and in control.

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